

DAY18:Exploring Python Built-in Modules & Practice Projects!

math

```
import math
print(math.pow(2,3))
print(math.ceil(2.3))
print(math.floor(2.3))
print(math.sqrt(16))
print(math.sin(9))
print(math.pi)
print(math.cos(9))
```

random

```
import random
print(random.randint(1000,9999))
print(random.randrange(1,100))
print(random.choice(['python','java','c','c++']))
choice = ['python','java','c','c++']
random.shuffle(choice)
print(choice)
```

platform

```
import platform
print(platform.system())
print(platform.release())
print(platform.version())
print(platform.machine())
print(platform.processor())
print(platform.python_version())
```

collections

```
import collections
data = ['apple', 'banana', 'apple', 'orange', 'banana', 'apple']
counter = collections.Counter(data)
print(counter)
print(counter.most_common())
```

```
from collections import defaultdict
data = defaultdict(list)
data['fruits'].append('apple')
data['python'].append('java')
data['arjun'].append('reddy')
print(data)
```

date time

```
from datetime import datetime
now = datetime.now()
print(now)
print(now.year)
print(now.month)
print(now.day)
print(now.hour)
print(now.minute)
print(now.strftime("%Y-%m-%d %H:%M:%S"))
print(now.strftime("%A"))
print(now.strftime("%B"))
print(now.strftime("%I:%M %p"))
```

guess the number

```
import random
attempts = 3
number = random.randrange(1, 100)
print(number)
while attempts > 0:
    guess = int(input("Guess the number between 1 and 100:
"))
```

```

    if guess == number:
        print("Congratulations! You guessed the number.")
        break
    else:
        attpts -= 1
        print(f"Wrong guess. You have {attpts} attempts left.")
if attpts == 3:
    print('You won 500 rupees')
elif attpts == 2:
    print('You won 300 rupees')
elif attpts == 1:
    print('You won 100 rupees')
else:
    print("Sorry, you've run out of attempts. The number was:", number)

print("Thank you for playing!")

itertools
-----
import itertools
a=itertools.count(10)
print(next(a))

b=itertools.repeat(2,5)
for i in b:
    print(i)

c=itertools.cycle(['python','java','c','c++'])
for i in range(5):
    print(next(c))

import itertools
d=itertools.chain(['python','java'],['c','c++'])
print(set(d))
'''

```

